Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Junior software developer with 2+ years building **education platforms**, **relational data models**, validation workflows, REST integrations, role-based access controls, dashboards, and CI/CD releases across Python, JavaScript, React, and SQL.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed **normalized relational schemas and role-level data isolation** for an AI education platform supporting 500+ users, improving dashboard performance from 1.2 seconds to 350 milliseconds through SQL query and indexing optimization.
- Built **ETL-style integrations** across FastAPI, SQL Server, PostgreSQL, and cloud storage, processing 5,000+ monthly jobs with reconciliation checks, error handling, and CI/CD-backed releases achieving 99%+ reliability in production.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **React and Flask modules** for question uploads, rubric generation, grading, and history dashboards, translating instructor requirements into maintainable workflows serving 150+ users and 10,000+ daily requests.
- Implemented **secure PDF, DOCX, and ZIP imports** with validation, REST APIs, concurrent processing, and persistent results, reducing feedback turnaround 60% while supporting reliable academic grading operations at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Translated integration requirements into **OOP-based APIs and ETL pipelines** using AWS Glue, S3, and SQL, improving response times 30% and data-pipeline efficiency 68% across production workflows for users.
- Maintained **20+ unit and integration test suites** with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, strengthening release validation, code coverage reporting, and version-controlled delivery processes.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **eight microservices** for authentication, role-based access, customer records, billing, notifications, and resource workflows, using Azure SQL, REST APIs, and WebSockets to synchronize 2M+ records across business locations.
- Developed **React interfaces and Node.js APIs** for internal CRM workflows, then automated Docker packaging, testing, and staged Jenkins deployments, cutting release time 50% and reducing environment drift.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Designed **Flask APIs** for booking, pricing, coupon validation, payments, and real-time status workflows across 50,000+ records, maintaining sub-200ms performance through Firebase synchronization and structured request validation rules.
- Mapped **12+ third-party integration points**, presented end-to-end architecture to the client, and implemented payment webhooks, refund automation, and cross-platform workflows that reduced manual payment handling by 70%.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML, CSS, Bash
Web & APIs: React.js, Next.js, Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets, JSON
Data: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, relational modeling, ETL, validation, dashboards
Security & QA: RLS, RBAC, JWT, PyTest, Jest, Selenium, unit/integration testing, code coverage
Delivery: Git, GitHub, JIRA, Agile/Scrum, CI/CD, Jenkins, Docker, AWS, Azure, release management

Projects

HMIS Live | *TypeScript, MERN, ABDM APIs, Sentry*
- Built a **healthcare information platform** with consent-based record access, appointment workflows, TypeScript dashboards, ABDM API integrations, Twilio notifications, and Sentry monitoring, reducing administrative workload by 30% for staff.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL development workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified Java and TypeScript codebase with Docker-backed development workflows.

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*
- Built a **fintech web platform** supporting 50,000+ users and 1M+ transactions, combining Next.js filtering, Plaid bank linking, real-time updates, MongoDB access, and Kubernetes-based CI/CD deployments for resilient operations.